

Lesson 1: Statistics and Machine Learning



Lesson Overview

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1.1 Relevance of Statistics in Big Data and Machine Learning



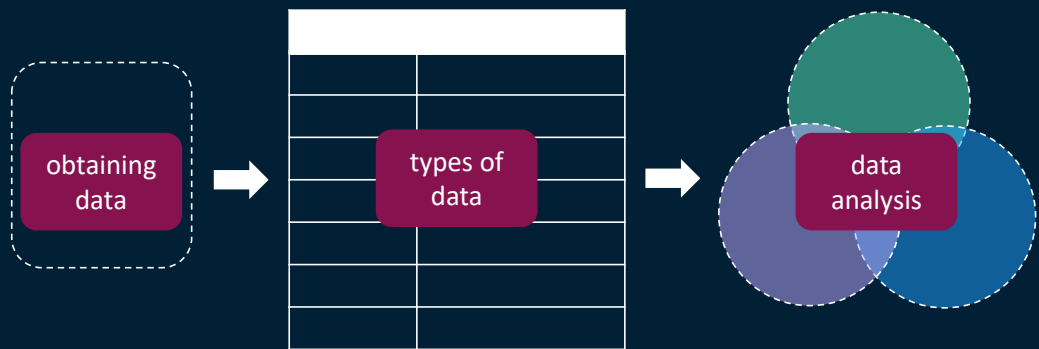
Data and Digital Economies

“ Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write. ”

– Samuel S Wilks (1951) *paraphrased from* Herbert G Wells (1903)

Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write! This is the quotation from the presidential address in 1951 of mathematical statistician Samuel S. Wilks to the American Statistical Association. Wilks was paraphrasing Herbert G. Wells from his book "Mankind in the Making" published in 1903.

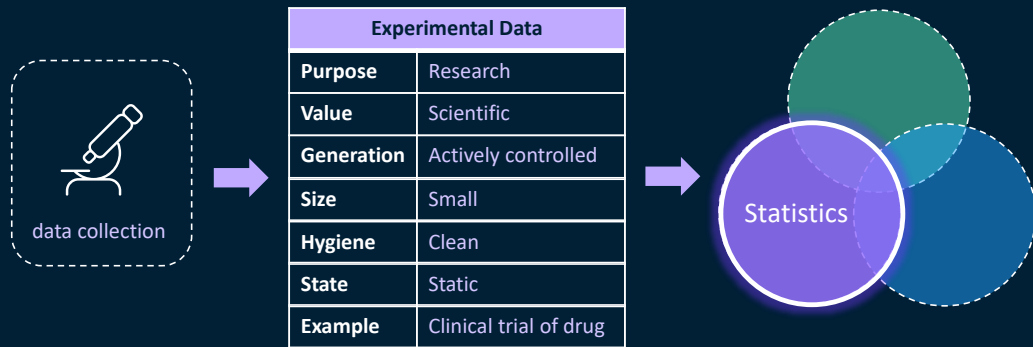
Data and Digital Economies



So, what's the relevance of statistics in the big data machine learning world? Before taking on this question, it's helpful to understand data and digital economies. Data are the foundation of everything you do.

First, let's look at the ways we obtain data, the different types of data, and how each type of data is analyzed.

Data and Digital Economies

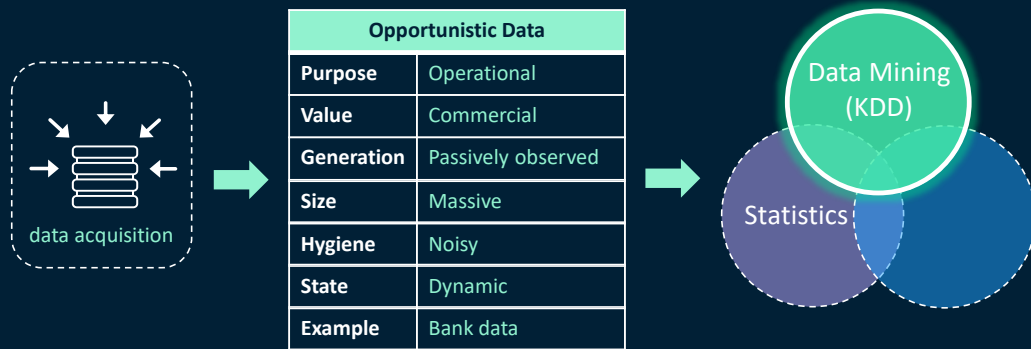


Statistics is the science of **collecting, analyzing, interpreting, and presenting** empirical data, and accounting for the relevant uncertainties.

Historically, most data were generated or collected for research purposes. Therefore, their value is scientific. This type of data is commonly called *experimental data*. Experimental data are actively controlled, so they are clean, small, and static. Examples of experimental data include a clinical trial of a drug and the effect of fertilizers on agriculture yield.

Such data analysis is performed using statistics. *Statistics* is the science of collecting, analyzing, interpreting, and presenting empirical data, and accounting for the relevant uncertainties. Traditional statistics relied heavily on small samples and rigid assumptions about data and its distributions. Large and complex data sets were rare - and those few that did exist were rarely analyzed.

Data and Digital Economies



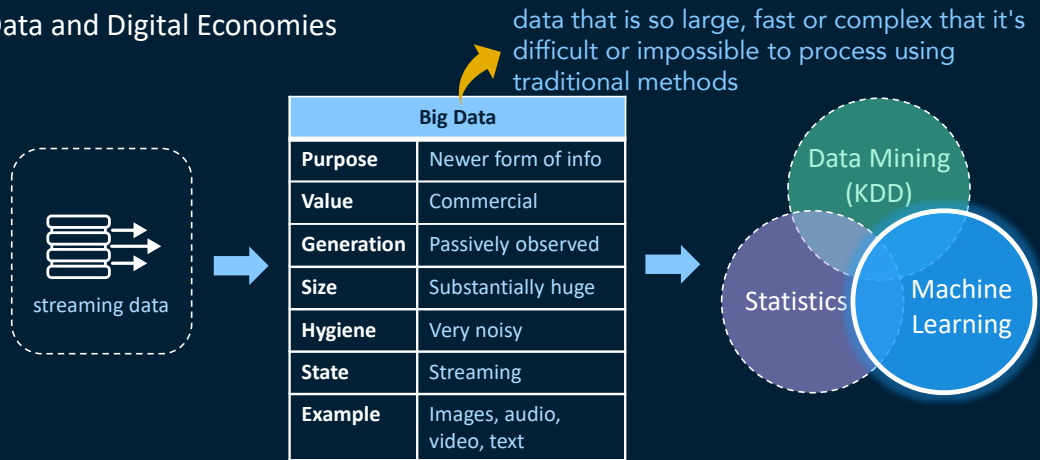
Data mining is the process of **sorting through large data sets to identify patterns and relationships** and **predict future trends** that can help solve business problems through data analysis.

opportunistic data (Huber 1997)

Over time, businesses accumulated massive amounts of operational data. This operational data was not generated with data analysis in mind. It is aptly characterized as *opportunistic data*. This contrasts with experimental data where factors are controlled and varied in order to answer specific questions. This type of data are collected in a *database* - an organized collection of structured information, or data, typically stored electronically in a computer system.

Opportunistic data are passively observed and are noisy, massive, and dynamic. An example is data coming from bank operations. Such data is analyzed using *data mining*. Data mining is the process of sorting through large data sets to identify patterns and relationships and predict future trends that can help solve business problems through data analysis. *Knowledge discovery in databases*, abbreviated as KDD, is often used synonymously with data mining. KDD is a multidisciplinary research area that uses statistics, in the form of exploratory data analysis and predictive models, to reveal patterns and trends in very large data sets.

Data and Digital Economies



Machine learning uses many statistics and data mining concepts and methods to automate model building that can **learn iteratively** from data with minimal human intervention.

Now, newer forms of data are coming in. We've moved *away* from systems of inconvenient data acquisition - requiring days, weeks, or months - and we've moved *to* streaming data. *Streaming data* is data that is continuously generated by different sources, such as social media, sensors, and cameras. Such data arrives at exceptional speed, in a variety of forms, and in humongous volumes. Everyone might have access but might not have the skills to make sense out of the data.

This large volume of data – both structured and unstructured – inundates a business on a day-to-day basis. Big data refers to data that is so large, fast, or complex that it's difficult or impossible to process using traditional methods.

To analyze this type of data, we must use and develop computer systems that can learn and adapt without following explicit instructions, by using algorithms and statistical models to analyze and draw inferences from patterns in data.

Machine learning uses many statistics and data mining concepts and methods to automate model building that can learn iteratively from data, identify patterns, and predict future results – with minimal human intervention.

What Is Big Data?

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: Big Data

Which of the following are examples of big data? (Select all that apply.)

- a. Sample of controlled clinical drug trials
- b. Social media feeds
- c. Feeds coming in from cameras installed on city roads
- d. Taking two samples of the same plant and exposing one of them to sunlight, and the other is kept away from sunlight

Answer: Big Data

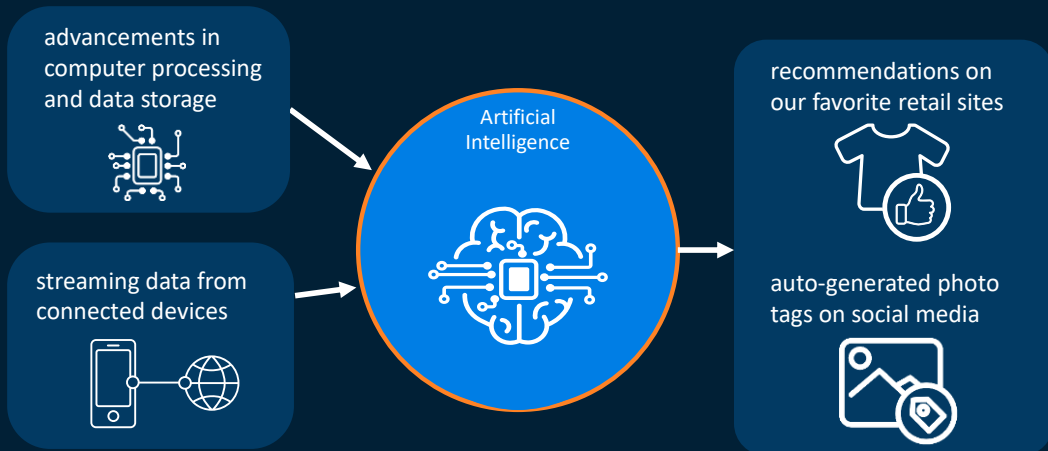
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- ☒ b. Social media feeds
- ☒ c. Feeds coming in from cameras installed on city roads
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Correct answer: b and c.

Note that **a** and **d** are examples of experimental data.

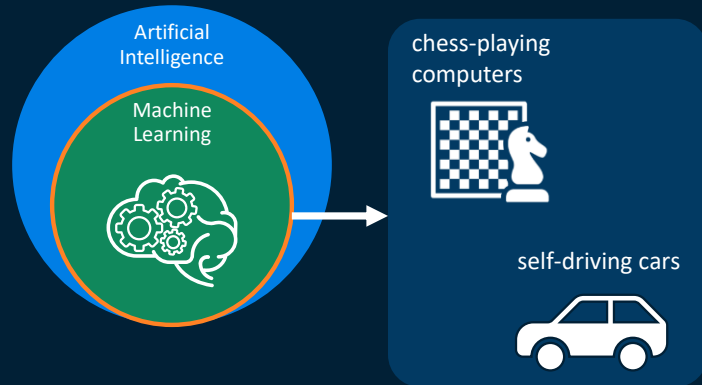
Big Data Produced Smart Applications



Advancements in computer processing and data storage made it possible to ingest and analyze more data than ever before. Around the same time, we started producing more and more data by connecting more devices and machines to the internet and streaming large amounts of data from those devices.

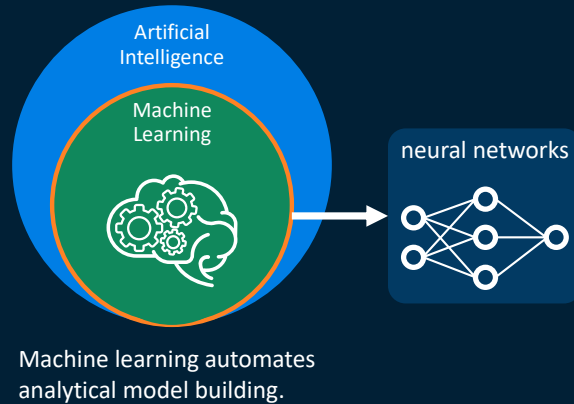
All these advancements brought artificial intelligence closer to its original goal of creating intelligent machines, which we're starting to see more and more in our everyday lives. From recommendations on our favorite retail sites to auto-generated photo tags on social media, many common online conveniences are powered by artificial intelligence.

Big Data Produced Smart Applications



Most artificial intelligence examples that you hear about today – from chess-playing computers to self-driving cars – rely heavily on machine learning and deep learning.

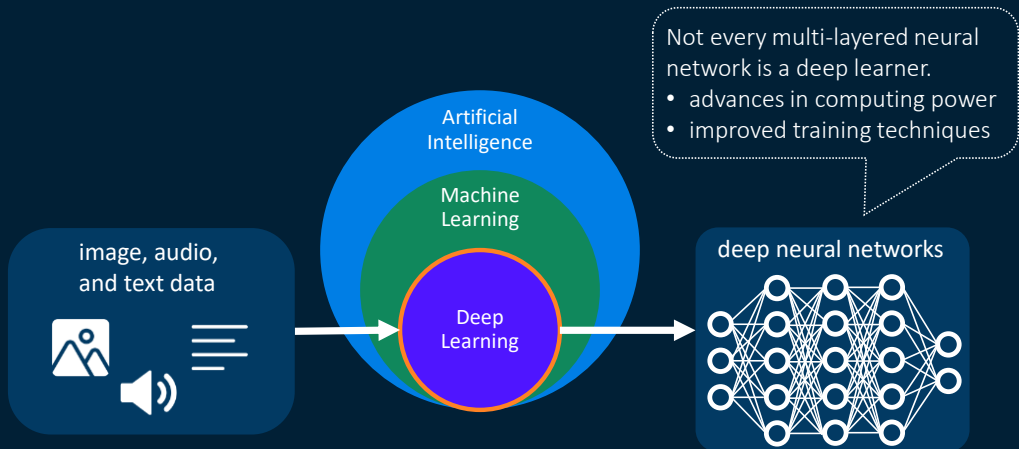
Big Data Produced Smart Applications



Artificial intelligence is the broad science of mimicking human abilities, and machine learning is a specific subset of artificial intelligence that trains a machine how to learn. Machine learning automates analytical model building. It uses methods from neural networks, statistics, operations research, and physics to find hidden insights in data without being explicitly programmed where to look or what to conclude.

Early work with neural networks during 1950s-1970s stirred excitement for "thinking machines." A neural network is a kind of machine learning inspired by the workings of the human brain. It's a computing system made up of interconnected units (like neurons) that processes information by responding to external inputs, relaying information between each unit. The process requires multiple passes at the data to find connections and derive meaning from undefined data. Machine learning became popular in 1980s-2010s.

Big Data Produced Smart Applications

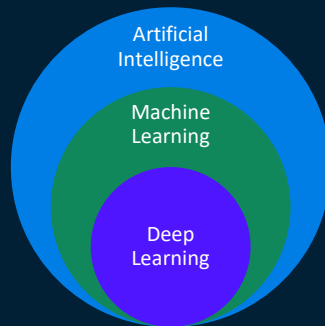


Machine learning and deep learning are subfields of artificial intelligence, and deep learning is a specific field of machine learning. While machine learning mostly dealt with huge tabular data, deep learning generally dealt with image, audio and text data that results in even larger tables.

Recent deep learning breakthroughs are driving an AI boom. *Deep learning* uses huge neural networks with many layers of processing units to learn complex patterns in large amounts of data.

Does this mean that every multi-layered neural network is a deep learner? No. Although the term deep learning refers to the numerous hidden layers used in a neural network, they take advantage of advances in computing power such as GPUs and improved training techniques such as better performing activation functions, randomly initialized weights, advanced regularization techniques, and efficient optimization algorithms.

Big Data Produced Smart Applications



Common Uses of AI

computer vision



natural language processing



internet of things



Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world. Using digital images from cameras and videos and deep learning models, machines can accurately identify and classify objects — and then react to what they "see." When machines can process, analyze, and understand images, they can capture images or videos in real time and interpret their surroundings. From recognizing faces to processing the live action of a football game, computer vision rivals and surpasses human visual abilities in many areas.

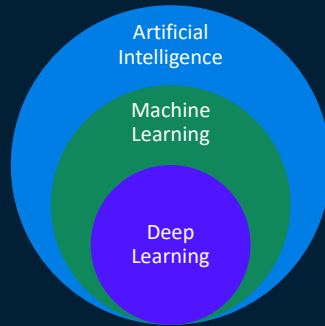
Natural language processing, or *NLP*, is a branch of artificial intelligence that helps computers understand, interpret, and manipulate and generate human language, including speech.

NLP helps computers communicate with humans in their own language, making it possible for computers to read text, hear speech, interpret it, measure sentiment, and determine which parts are important. The next stage of NLP is natural language interaction, which allows humans to communicate with computers using normal, everyday language to perform tasks.

The Internet of Things, or *IoT*, refers to a vast number of "things" that are connected to the internet so that they can share data with other things — IoT applications, connected devices, industrial machines, and more. Internet-connected devices use built-in sensors to collect data and, in some cases, act on it. IoT connected devices and machines can improve how we work and live. Real-world Internet of Things examples range from a smart home that automatically adjusts heating and lighting to a smart factory that monitors industrial machines to look for problems, and then automatically adjusts to avoid failures. The Internet of Things generates massive amounts of data from connected devices, most of it unanalyzed. Automating models with AI will allow us to use more of it.

Machine learning is based on the idea that machines should be able to learn and adapt through experience, and AI refers to a broader idea where machines can execute tasks "smartly." Artificial Intelligence applies machine learning, deep learning, and other techniques to solve actual problems.

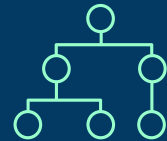
Big Data Produced Smart Applications



technologies that enable and support AI



graphical processing units (GPUs)



advanced algorithms



application programming interfaces (APIs)

In addition, several technologies enable and support AI.

Graphical processing units, or *GPUs*, are key to AI because they provide the heavy compute power that's required for iterative processing. Training neural networks requires big data plus compute power.

Advanced algorithms are being developed and combined in new ways to analyze more data faster and at multiple levels. This intelligent processing is key to identifying and predicting rare events, understanding complex systems, and optimizing unique scenarios.

Application programming interfaces, or *APIs*, are portable packages of code that make it possible to add AI functionality to existing products and software packages. They can add image recognition capabilities to home security systems and Q&A capabilities that describe data, create captions and headlines, or call out interesting patterns and insights in data.

In summary, the goal of AI is to provide software that can reason on input and explain on output. AI will provide human-like interactions with software and offer decision support for specific tasks, but it's not a replacement for humans – and won't be anytime soon.

Question: Computer Vision

Which of the following is an example of computer vision?

- a. Facial recognition-based security access in your phone
- b. Smart watch connected to your phone
- c. Siri or Google Assistant in your phone
- d. Text autocorrect feature in your phone

Answer: Computer Vision

Which of the following is an example of computer vision?

- ☒ a. Facial recognition-based security access in your phone
- b. Smart watch connected to your phone
- c. Siri or Google Assistant in your phone
- d. Text autocorrect feature in your phone

Correct answer: a.

Note that **b** is an example of Internet of Things (IoT) and **c** and **d** are examples of natural language processing (NLP).

Knowing Statistics (1)

What do you think.... To do machine learning on big data, do you need to know statistics?

- ☐ No, not at all.
- ☐ Might be, quite unsure.
- ☐ Yes, definitely.



Let's explore running
an automated
machine learning
task.

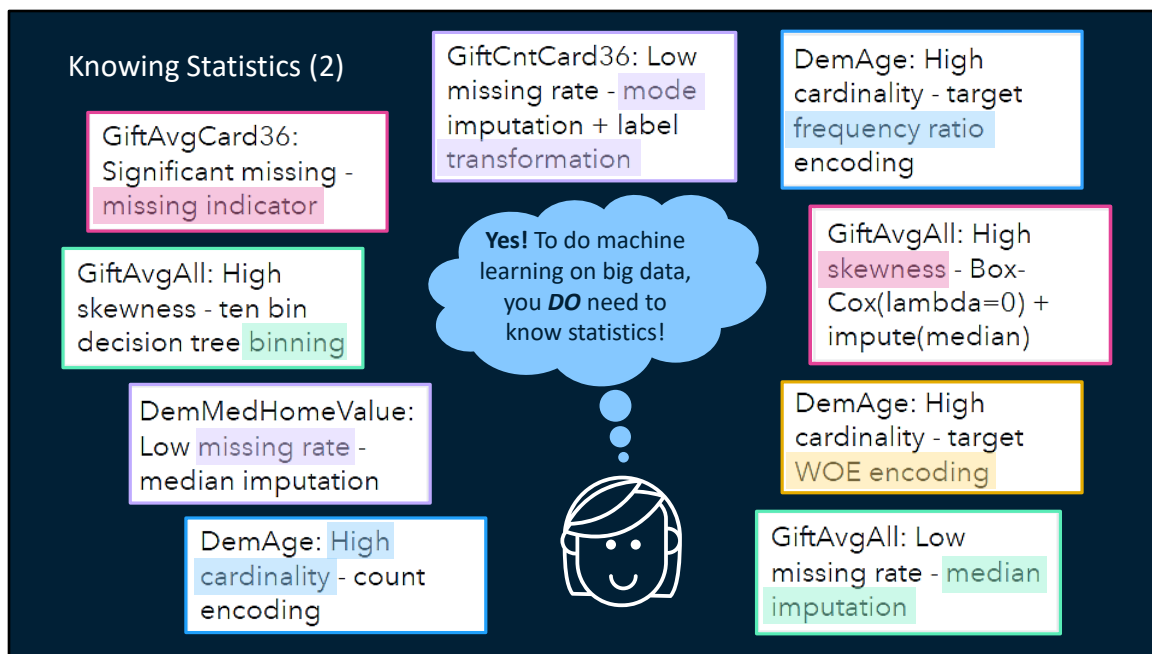


To do machine learning on big data, do you need to know statistics? At this point, you might not be sure. To answer this question, let's explore running an automated machine learning task.

Demo: Automated Machine Learning in SAS Studio

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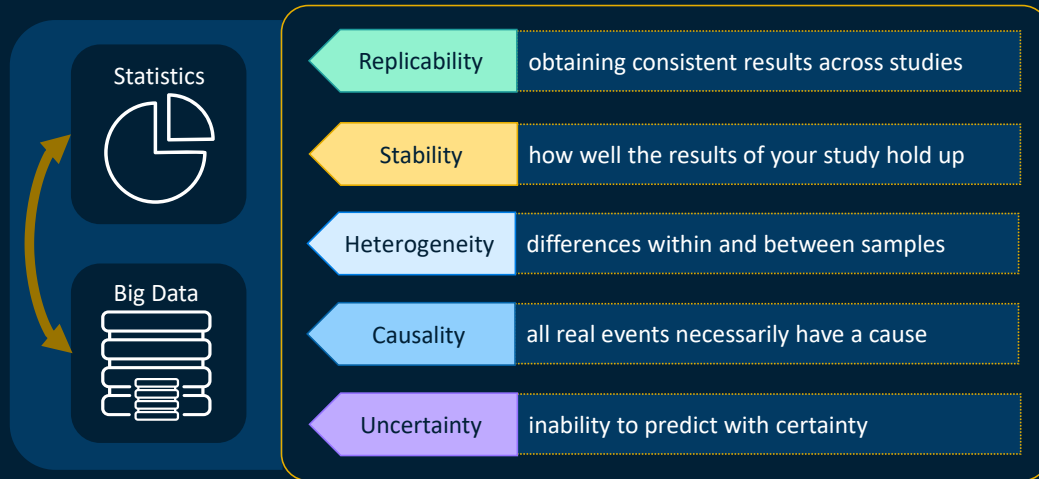




In the previous demonstration, even though the machine learning capability in the tool can fulfill your objective of creating new features automatically, you still need to have the knowledge of basic statistical concepts to understand and explain the output better. You encountered some of the statistical terms highlighted here, for example, missing indicator, mode, transformation, frequency ratio, binning, skewness, missing rate, WOE encoding, high cardinality, median imputation, and so on. All of these and more will be discussed in this course. Yes! To do machine learning on big data, you do need to know statistics!

Machine learning tools use **data-driven algorithms** and **statistical methods** to prepare and analyze data sets and then **draw inferences** from identified patterns or **make predictions** based on them.

Relevance of Statistics in Big Data



statistics for big data (Bühlmann and Geer 2018)

Statistics remains highly relevant irrespective of the "bigness" of data. The role of statistics remains what it has always been, and is even more important now. Perhaps the core statistical task in (traditional) statistics is inductive inference from data to models and scientific conclusions. This core task is still very relevant in the advent of massive data sets.

Replicability, stability, heterogeneity, causality, and uncertainty are the five basic principles of statistics, and they all hold equally well with big data.

Replicability is obtaining consistent results across studies aimed at answering the same scientific question, each of which has obtained its own data.

Related to replicability is the notion of stability. Stability is how well the results of your study or experiment consistently hold up - in other words, the strength of support for your results subject to small perturbations in input data. It's a useful concept to address whether an alternative, "appropriate" analysis would generate similar knowledge. More specifically, it's a measure of how well you control for random errors in your study.

Heterogeneity is a vital concept that appears in various contexts, and its definition varies accordingly. Heterogeneity can indicate differences within individual samples, between samples, and between experimental results in a meta-analysis.

For a simple causation definition, statistics describes a relationship between two events or two variables. Causation is present when the value of one variable or event increases or decreases as a direct result of the presence or lack of another variable or event.

The uncertainty principle implies that it is in general not possible to predict the value of a quantity with arbitrary certainty, even if all initial conditions are specified.

Ideally, in big data scenario too, the conclusions and findings are **replicable** and generalizable. If you imagine running the analysis again, now on a new data set, would the outcome be similar, meaning that the model is **stable**? How would you find out what similarity in outcomes means and how to evaluate accuracy, to quantify **uncertainty**. Understanding **heterogeneity** in large-scale data sets is more crucial, and comprehending **causality** and its connection to robust prediction is still interesting.

More about the Relevance of Statistics in Big Data

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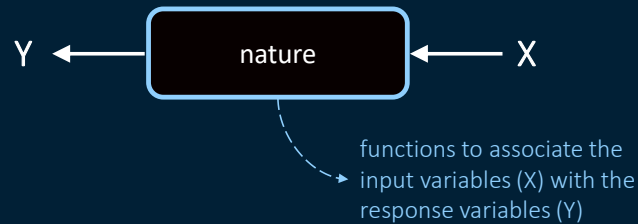
Transitioning to the Machine Learning World

“ There are **two cultures** in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown. ”

– Leo Breiman (2001)

According to Leo Breiman, "There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown." In other words, there is a need to transition from traditional statistical modeling to the machine learning world.

Analyzing Data



Explanation

How is nature associating the response variables to the input variables?



Prediction

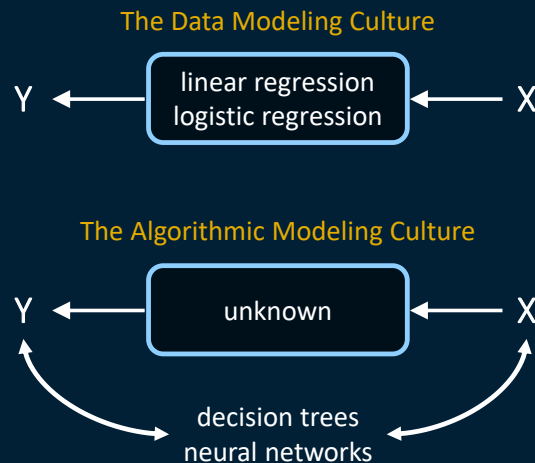
What will be the responses to future input variables?

Statistics starts with data. Think of the data as being generated by a black box in which a vector of input variables X go in one side, and on the other side, the response variables Y come out. Inside the black box, nature functions to associate the input variables with the response variables.

There are two goals in analyzing the data: explanation and prediction.

Explanation (or explanatory analysis) is the extraction of some information about how nature associates the response variables to the input variables. Prediction (or predictive modeling) refers to predicting the responses to future input variables. You will learn about both of these data analysis goals in the upcoming lessons.

Statistical Modeling: The Two Cultures



There are two different approaches toward these goals: the data modeling culture and the algorithmic modeling culture.

The analysis in the *data modeling culture* starts with assuming a stochastic data model for the inside of the black box, for example, a linear or a logistic regression model. The values of the parameters are estimated from the data and the model then used for information and/or prediction. Thus, the black box is filled in like the one shown at the top. Here the model is evaluated like a yes-no goodness-of-fit tests and residual examination.

The analysis in the *algorithmic modeling culture* considers the inside of the box complex and unknown. Their approach is to find a framework or mathematical function that operates on X to predict the responses Y . Their black box looks like the one shown at the bottom. In this culture, model is evaluated by predictive accuracy. Most of the data mining and machine learning models falls in this culture, for example, decision trees and neural networks. The difference with classical statistics is that they are less model based, but rather algorithm based, and data driven.

Statistical Modeling: The Two Cultures

marketing
campaign example



The Data Modeling Culture

What is the psychology of response behavior? Or, why a person with certain characteristics is not a responder?

The Algorithmic Modeling Culture

Whether a person is going to respond to a marketing campaign or not?

Thus, there are several things you might aim at. For example, consider a marketing campaign scenario. If your aim is to know whether a person is going to respond to a marketing campaign, you run an algorithm on his or her data, and out comes an answer. The main point is that the answer should typically be correct. You might not be so much interested in how the algorithm works. You even might be not interested in the psychology of behavior and why a person with certain characteristics is not a responder. This is a distinctive characteristic of a machine learning algorithm.

Question: Modeling Cultures

Which of the two modeling cultures do you think is used more often in the machine learning world?

- a. The data modeling culture
- b. The algorithmic modeling culture

Answer: Modeling Cultures

Which of the two modeling cultures do you think is used more often in the machine learning world?

- a. The data modeling culture
- ☒ b. The algorithmic modeling culture

Correct answer: b. In the algorithmic modeling culture, the model is evaluated by predictive accuracy. You might not be so interested in how the algorithm works. Most of the data mining and machine learning models fall in this culture.

Modeling Approaches

Traditional Approach

(SEMMA)

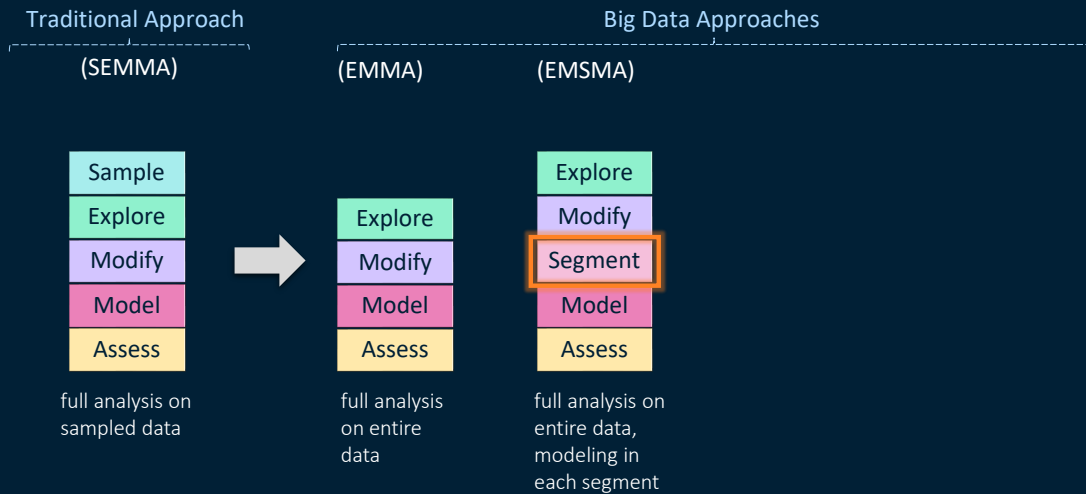


full analysis on
sampled data

Over time, SAS has followed a number of approaches for data mining and machine learning. The traditional approach is *SEMMA*, which stands for Sample, Explore, Modify, Model, and Assess.

First, you **sample** the data by creating one or more data tables. The samples should be large enough to contain the significant information, but small enough to process. Next, you **explore** the data by searching for anticipated relationships, unanticipated trends, and anomalies in order to gain understanding and ideas. Then you **modify** the data by creating, selecting, and transforming the variables to focus the model selection process. Next, you **model** the data by using the analytical tools to search for a combination of the variables that reliably predicts a desired outcome. Finally, you **assess** competing predictive models.

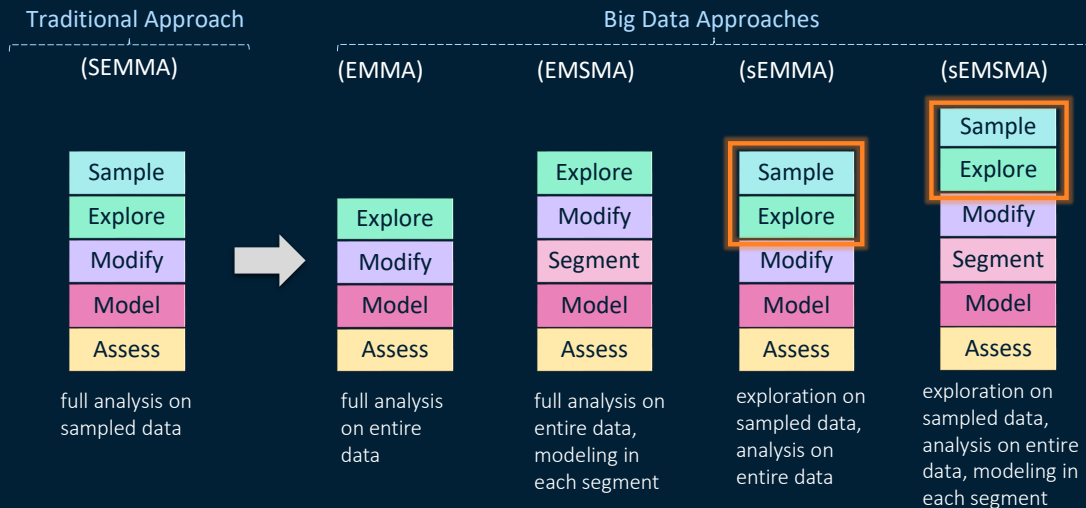
Modeling Approaches



With the advent of big data, this framework can be slightly modified to many frameworks. For example, *EMMA*: Explore, Modify, Model, and Assess. In *EMMA*, complete analysis is performed on the entire data, so sampling is not required.

Another variation, *EMSMA*, adds a Segment piece: Explore, Modify, *Segment*, Model, and Assess. Why would you add Segment? Big data has greater diversity in the sense that it represents two or more populations. Different relationships between an input and the target might exist in different populations. For this purpose, you might want to first segment the data and then run a separate predictive model in each segment. However, this modification is not universally accepted and exact.

Modeling Approaches

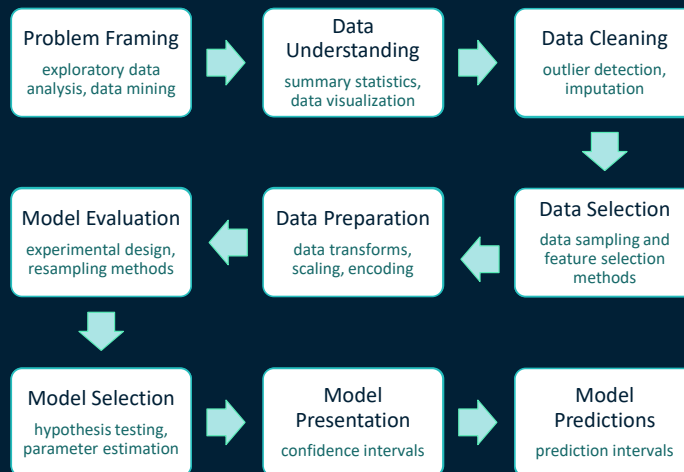


Sampling in SEMMA is optional, but generally used in case of big data. SAS Viya can handle big data, but it's not entirely correct to say that we don't need to sample now.

Why sample? Real life data is usually dirty and noisy because of inconsistencies, incompleteness, duplication, and merging problems. Preparing your data for analysis starts with exploring your data. Exploring the data can be time-consuming if processed using entire data. So, sampling is inevitable.

After gaining intimate knowledge of the variables by using both graphical and numerical methods, then maybe open the flood gates... let the full data flow... and develop your final, "deeply trained" model. In a big data world, sampling is, in fact, a **best practice** when it comes to rapidly learning and gaining insights, and this leads to modeling approaches where sampling is optional: *sEMMA* (sample, Explore, Modify, Model, Assess) and *sEMSMA* (sample, Explore, Modify, Segment, Model, Assess).

Why Is Statistics Important to Machine Learning?



Statistics is a pillar of machine learning. You cannot develop a deep understanding and application of machine learning without statistics. Let's discuss some typical steps where statistical methods are used in an applied machine learning project.

Problem Framing requires the use of exploratory data analysis and data mining.

Data Understanding requires the use of summary statistics and data visualization.

Data Cleaning requires the use of outlier detection, imputation and more.

Data Selection requires the use of data sampling and feature selection methods.

Data Preparation requires the use of data transforms, scaling, encoding, and much more.

Model Evaluation requires experimental design and resampling methods so that you have the right data to effectively answer your analytical questions.

Model Selection requires the use of statistical hypothesis tests and estimation statistics.

Model Presentation requires the use of estimation statistics such as confidence intervals.

Model Predictions requires the use of estimation statistics such as prediction intervals.

Jason Brownlee discusses these topics in detail in his book **Statistical Methods for Machine Learning**.

Journey from Statistics to Machine Learning

Traditional Statistics	Machine Learning
• Designed for inference about the relationships between variables.	• Designed to make the most accurate predictions possible.
• A mathematical perspective on modeling data, focusing on data models and on goodness of fit.	• A computer science perspective on modeling data, focusing on algorithmic methods and model skill.
• Prediction comes last.	• Prediction is most important.
• Relies heavily on small samples and assumptions about data and distributions.	• Historically, relies heavily on computing power.

When it comes to using data, there are two main camps: traditional statistics and machine learning. The major difference between them is their purpose. Statistical models are designed for inference about the relationships between variables. Machine learning models are designed to make the most accurate predictions possible.

You already know that these two complement each other when it comes to analyzing data. *Statistics* and *statistical learning* are a mathematical perspective on modeling data with a focus on data models and on goodness of fit. *Machine learning* and *predictive modeling* are a computer science perspective on modeling data with a focus on algorithmic methods and model skill.

Much traditional statistics is motivated primarily by showing that one factor causes another, for example, clinical trials. Understanding comes next, prediction last. In machine learning and data mining, the order is usually reversed -- prediction is most important.

Traditional statistical techniques were mostly developed where computing power was not an option. As a result, traditional statistics relies heavily on small samples and heavy assumptions about data and its distributions. In contrast, historically, machine learning techniques and approach heavily rely on computing power.

More about the Journey from Statistics to Machine Learning

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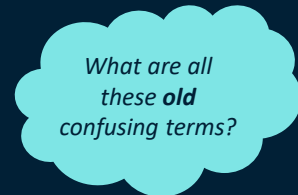
1.2 Terminology and Vocabulary



Basic Terminology



Statistician



Data Scientist

Analytical terms might have different meanings in different contexts, specifically statistics versus machine learning (ML). There are many words for the same thing, and one word might have different meanings to different users. For example, a statistician might say: "*What are all these new fangled and confusing terms?*" Similarly, a data scientist might feel: "*What are all these archaic, outmoded, and confusing terms?*" Always be sure to clarify with your audience that they are using the same definition that you are, and that everyone understands the terms being used.

Basic Terminology

statistical
terms

machine learning
terms

What are all
these **new**
confusing terms?



Statistician

Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...

- Data set
- Data table

What are all
these **old**
confusing terms?



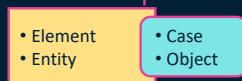
Data Scientist

Consider a simple data set containing information on age, gender, working hours per week and income. A *data set* is a table of data from which the machine learns. Let's get acquainted with basic analytical terminology and see some examples of synonyms.

Statistical terms are shown in yellow boxes and their equivalent machine learning terms are shown in the aqua boxes.

Basic Terminology

Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...



The elements are the entities on which data are collected or extracted. In machine learning terminology, they are called cases or objects. John, Smith and Emma are cases here.

Basic Terminology

Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...

- Element
- Entity

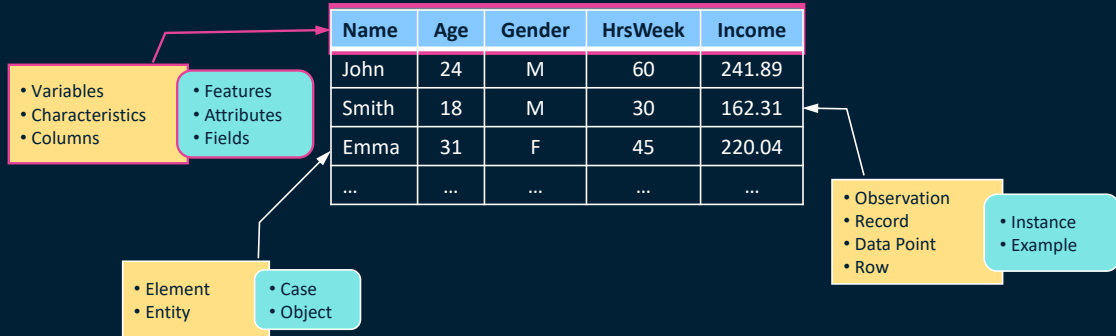
- Case
- Object

- Observation
- Record
- Data Point
- Row

- Instance
- Example

The set of measurements collected or extracted for a particular element is called an *observation* or a *record* or a *data point*. Due to specific format of analysis-based tables, at times it is also called a *row*. In machine learning terminology, it is known as an *instance* or an *example*. The highlighted row is an instance here.

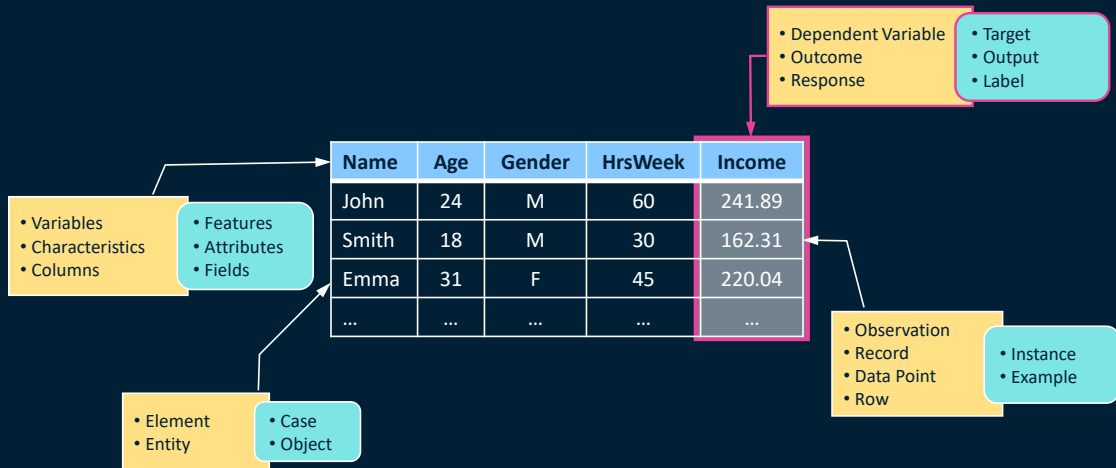
Basic Terminology



A *variable* is a *characteristic* of interest for the elements. Due to specific format of analysis base table, at times, they are also called *columns*. The total number of data values in a data set is the number of elements multiplied by the number of variables.

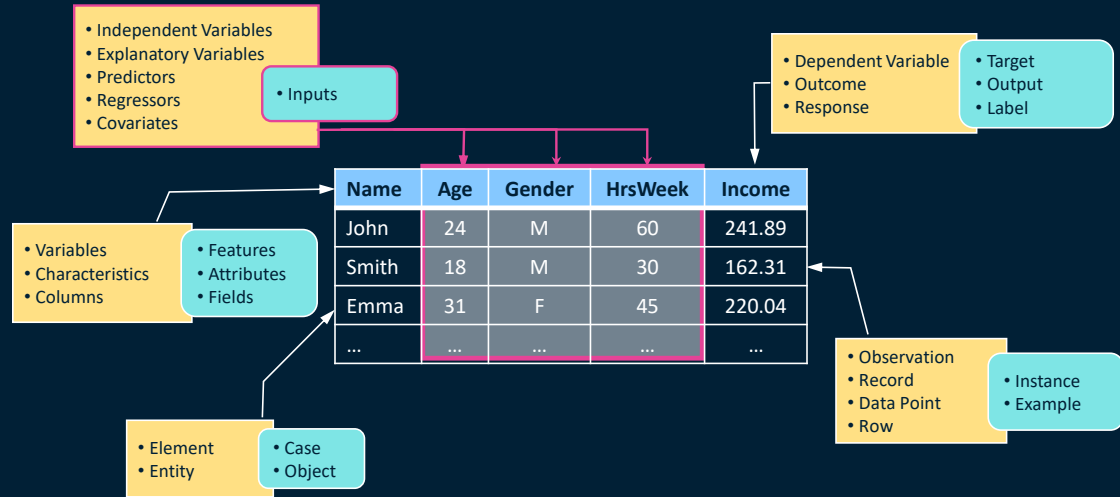
Similarly, an *attribute* or a *field* is an aspect of an instance. For example, age, gender, hours per week, and income are variables. Attributes are often called features in machine learning.

Basic Terminology



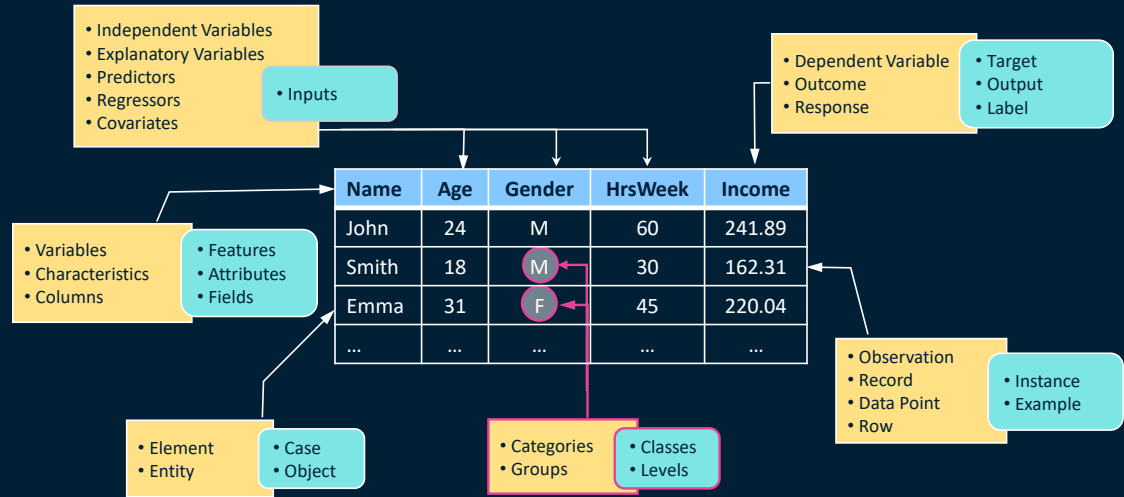
Dependent variable receives this name because, in an experiment, its values are studied under the supposition that they depend on the values of other variables. For example, how much you earn depends on your age and hours per week. In traditional statistical terms, it is also known as *outcome* or *response*. It is sometimes called a variable whose value need to be predicted. In machine learning terminology, it is simply called a *target* or *output*. In case of a categorical target, it can be called a *label*. It is seldom called a training value in data science.

Basic Terminology



Independent variables, in turn, are not seen as depending on any other variable in the scope of the experiment in question. Independent variables are variables that stand on their own, and you have complete control over which independent variables you choose. During an experiment, you usually choose independent variables that you think will affect dependent variable. For example, age and hours per week might affect the income, but not gender. In traditional statistical terms, they are also known as *explanatory variables* or *predictors* or *regressors* or *covariates*. In machine learning terminology, they are simply called *inputs*.

Basic Terminology



In this example, the variable gender is somewhat different from variables: age, hours per week, and income. Gender does not have fractional values, rather it observes only three values – male, female, and others. They are called *categories* or *groups* in statistical terminology or *classes* or *levels* in the machine learning world. Discussed next is the difference between continuous and categorical variables.

Question: Basic Terminology (1)

Which of the following pairs of terms define the highlighted part in the data table? (Select all that apply.)

- a. Target / Response
- b. Instance / Example
- c. Predictor / Input
- d. Observation / Record

Company	Stock Exchange	Annual Sales (\$M)	Earning per Share (\$)
MedTech	AMEX	84.20	0.33
EnergyWest	NYSE	372.85	1.89
DataDig	BSE	74.65	2.10
...	...	...	...

Answer: Basic Terminology (1)

Which of the following pairs of terms define the highlighted part in the data table? (Select all that apply.)

- a. Target / Response
- ☒ b. Instance / Example
- c. Predictor / Input
- ☒ d. Observation / Record

Company	Stock Exchange	Annual Sales (\$M)	Earning per Share (\$)
MedTech	AMEX	84.20	0.33
EnergyWest	NYSE	372.85	1.89
DataDig	BSE	74.65	2.10
...	...	...	...

Correct answer: b and d.

The highlighted row - a set of measurements collected or extracted for EnergyWest company - is an *observation* or a *record*. In machine learning terminology, it is known as an *instance* or an *example*.

Note that the terms in **a** are equivalent for the outcome variable or the dependent variable. The terms in **c** are equivalent for the independent variable also known as explanatory variable.

Question: Basic Terminology (2)

Which of the following pairs of terms define the highlighted part in the data table? (Select all that apply.)

- a. Variable / Characteristic
- b. Category / Group
- c. Class / Level
- d. Feature / Attribute

Company	Stock Exchange	Annual Sales (\$M)	Earning per Share (\$)
MedTech	AMEX	84.20	0.33
EnergyWest	NYSE	372.85	1.89
DataDig	BSE	74.65	2.10
...	...	...	...

Answer: Basic Terminology (2)

Which of the following pairs of terms define the highlighted part in the data table? (Select all that apply.)

- ☒ a. Variable / Characteristic
- ☐ b. Category / Group
- ☐ c. Class / Level
- ☒ d. Feature / Attribute

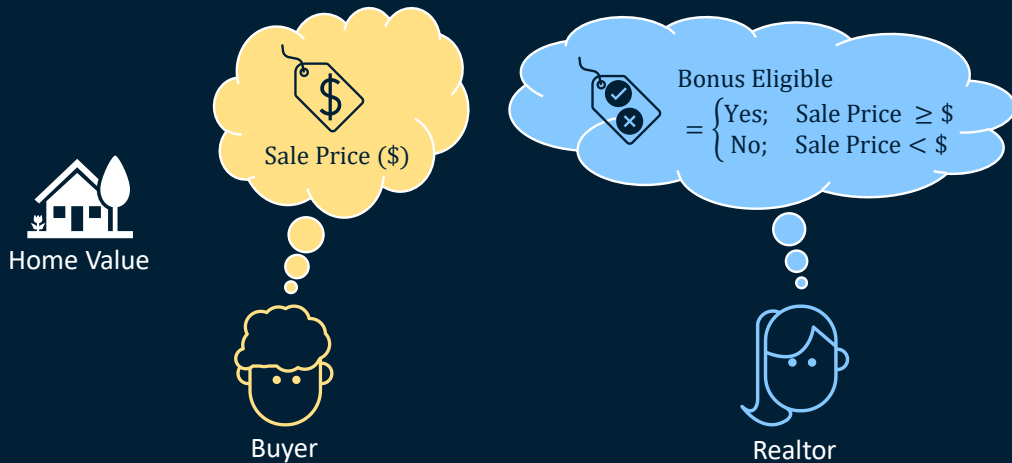
Company	Stock Exchange	Annual Sales (\$M)	Earning per Share (\$)
MedTech	AMEX	84.20	0.33
EnergyWest	NYSE	372.85	1.89
DataDig	BSE	74.65	2.10
...	...	...	...

Correct answer: **a** and **d**.

The highlighted column is a *variable* or a *characteristic* of interest. Similarly, an *attribute* is an aspect of an instance. Attributes are often called features in machine learning.

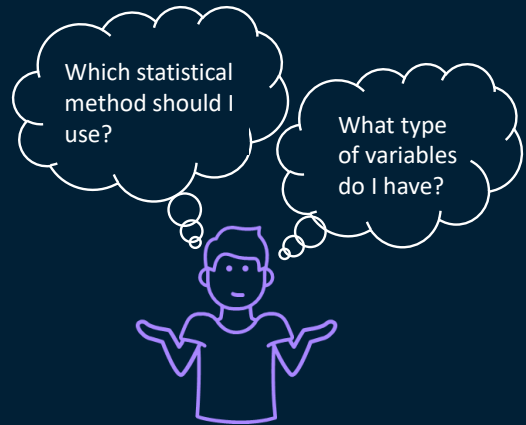
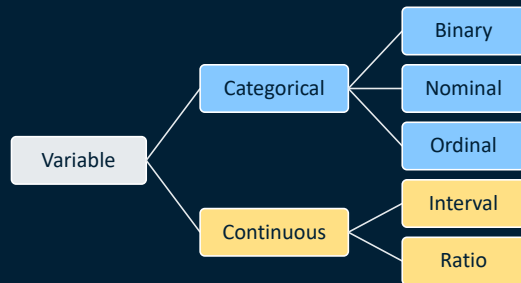
Note that the terms in **b** and **c** are equivalent. They are used to represent the variable values that have observed values rather than fractional values. Examples are the Stock Exchange values of AMEX, NYSE, BSE, and so on. They are called *categories* or *groups* in statistical terminology or *classes* or *levels* in the machine learning world.

Variable Type and Level of Measurement



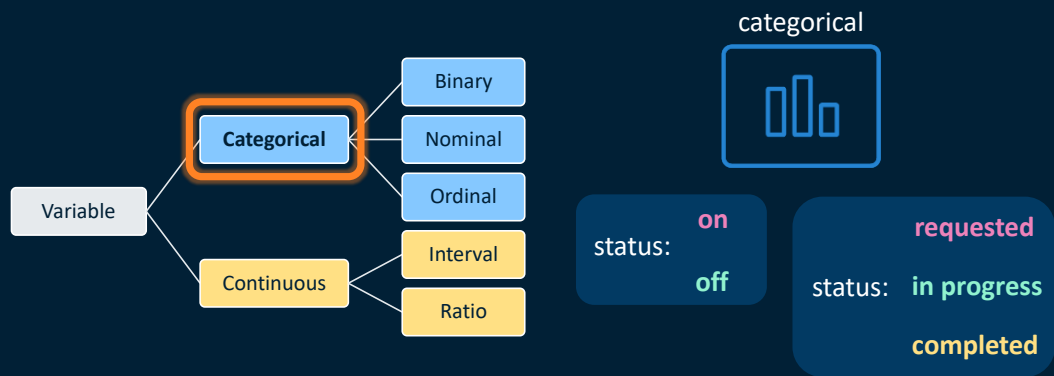
Data comes in several different types. For example, consider the sale of individual residential property. A buyer would be interested in exact sale price in dollars. So, the Sale Price variable will have many different dollar values. On the other hand, a realtor would be interested in considering the variable differently. Realtors receive some standard percentage of commission on homes sales if sale price is more than some dollar amount otherwise not. Thus, the same variable turned dichotomous, observing only two values: bonus eligible or not bonus eligible depending on dollar sale price.

Variable Type and Level of Measurement



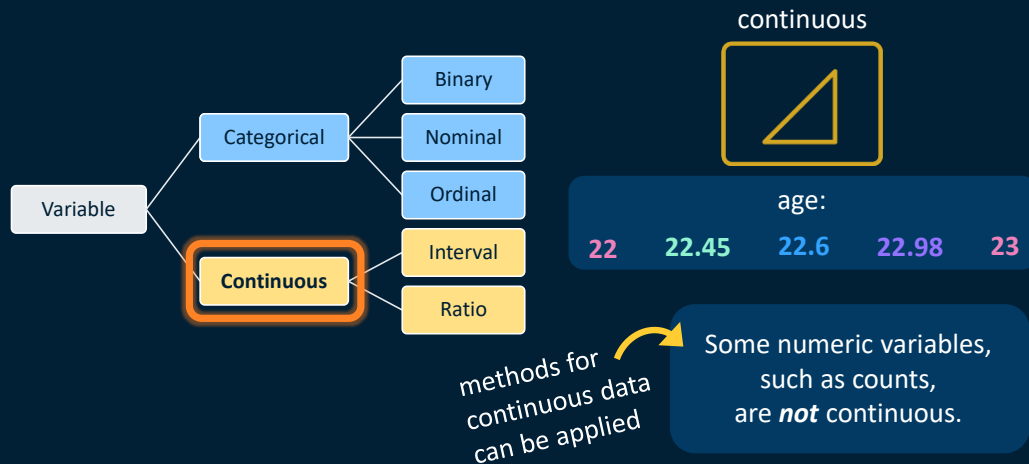
There are a variety of statistical methods for analyzing data. To choose the appropriate method, you must determine the type and level of measurement for your variables. The most basic distinction is that between continuous and categorical data, which has a profound impact on the types of statistics that can be used.

Variable Type and Level of Measurement



Categorical variables are variables that represent groupings. Categorical variables are also known as discrete or qualitative variables. Categorical variables can be stored as numeric or non-numeric values in SAS. Categorical variables can be further categorized as either binary, nominal, or ordinal.

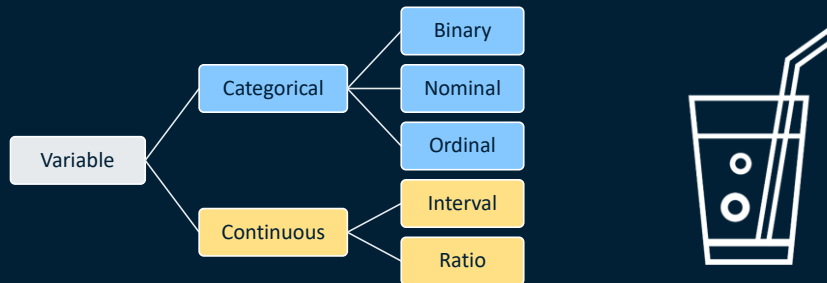
Variable Type and Level of Measurement



Continuous variables can take on any of an infinite number of possible values between two numbers, and the relative distances between the values are meaningful. An example of a continuous variable would be age. There is no restriction to the number of values between 22 and 23. You could have an age of 22.45 or 22.98 years. A variable can be continuous even if your measurement system has finite intervals. For instance, we normally report age as 22 years, 23 years, and so on.

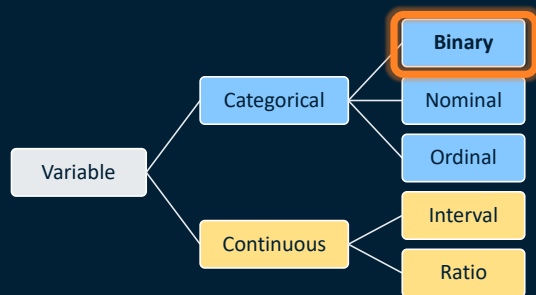
Some numeric variables are not continuous. These include variables such as counts that can only take on specific values (for example, integers). In many statistical applications, methods for continuous data can be applied to these variables as well provided there are many discrete values. Continuous variables can be either "ratio" level or "interval" level.

Variable Type and Level of Measurement



Suppose you have a certain beverage, say, a glass of soda. Keeping this as an object example, we can describe what is binary, nominal, ordinal, interval, and ratio scales.

Variable Type and Level of Measurement

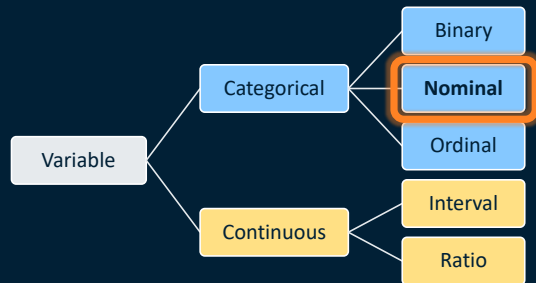


Variable: Do you have a beverage?
1 or 0 , Yes or No



Binary variables are variables that take only two values. For example, do you have a beverage or you do not have beverage. Many variables and questions are naturally binary, but it is often useful to construct binary variables from other types of data.

Variable Type and Level of Measurement

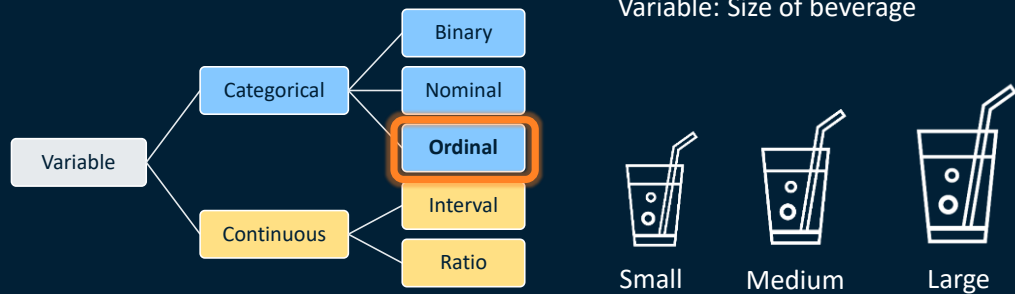


Variable: Type of beverage



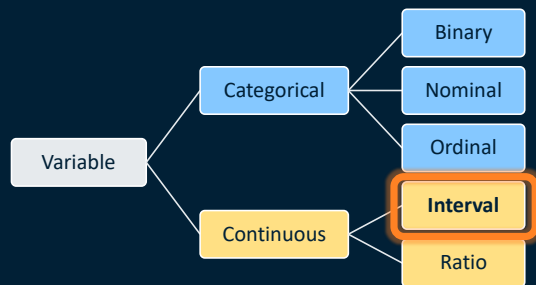
Data on a nominal scale represent a set of categories that differ in some characteristic that does not necessarily have a measurable magnitude. Observations on a nominal scale are classified in one of several non-overlapping categories, also called the *levels* of the variables. *Nominal variables* have values with no logical ordering. In this example, beverage type is a nominal variable, even though numeric values are assigned to the categories.

Variable Type and Level of Measurement

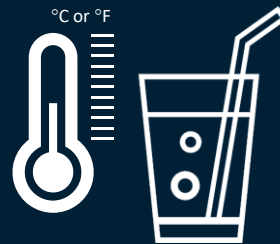


Ordinal variables have values with a logical order. However, the relative distances between the values are not clear. In this example, drink size is ordinal. The level assignments to the categories - small, medium, and large - convey information of the relative size of the drinks, but not precise information about the quantitative differences. For example, this scale did not suggest that small size was twice as small as the medium size nor that the difference between the small size and the medium size was the same as the difference between medium size and the large size beverage.

Variable Type and Level of Measurement

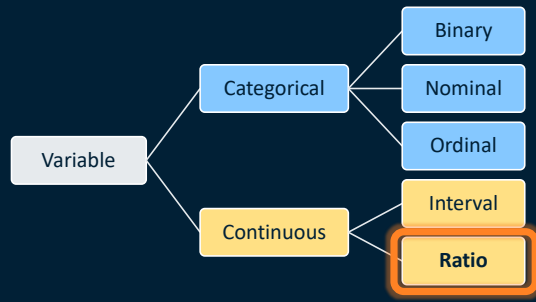


Variable: Temperature of beverage

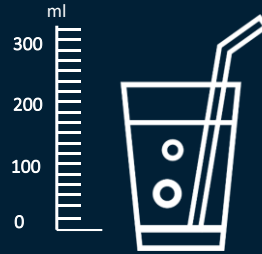


Interval measures have no true zero point, for example, temperature of beverage. Zero degrees Celsius does not imply the absence of heat. Interval scales are so called because when two measurements differ by an additive factor d , the larger is said to be d units higher than the lower. For example, if the temperature of your beverage is 40 degrees Fahrenheit and the temperature of another beverage is 30 degrees, then your drink is 10 degrees warmer, but it would be inaccurate to say that your drink is 25% hotter (as for data on a ratio scale). The difference between the volume in ml and the temperature in degrees Fahrenheit is that the Fahrenheit scale has an arbitrary 0 (at approximately -18 degrees Celsius), just as the Celsius scale has an arbitrary 0 at 32 degrees F.

Variable Type and Level of Measurement



Variable: Volume of beverage



👉 *SAS does not distinguish ratio variables from interval variables.*

Ratio measures have a true zero point, for example, volume of a beverage. Zero milliliters implies the absence of beverage. Ratio variables differ from that of interval variable in the existence of a "true" zero point. This zero point allows ratios to be calculated. They are called ratio scales because when two measurements differ by a multiplicative factor r , then the one value is said to be r times larger or smaller than the other. For example, if the volume of a drink is 100 ml and subsequently you poured 50 ml more in it, you can say that the volume has increased by 50%. The ratio factor is $r=1.5$. Likewise, a rise from 100 ml to 200 ml would represent a doubling in volume.

In SAS, ratio variables are not distinguished from interval variables, because the analytical methods used for them are the same.

Question: Measurement Scales

Which of the following represents a nominal measurement scale?

a. Jersey Numbers Assigned to Runners			
b. Rank of Winners	 Third place	 Second place	 First place
c. Performance Rating on a 0 to 10 Scale	8.2	9.1	9.6
d. Time to Finish, in Seconds	15.2	14.1	13.4

Answer: Measurement Scales

Which of the following represents a nominal measurement scale?

a. Jersey Numbers Assigned to Runners			
b. Rank of Winners	 Third place	 Second place	 First place
c. Performance Rating on a 0 to 10 Scale	8.2	9.1	9.6
d. Time to Finish, in Seconds	15.2	14.1	13.4

The correct answer is a. Rank of winners = ordinal. Performance rating on a 0-10 scale = interval. Time to finish in seconds = ratio.

Modeling Vocabulary

Analysis Data / Training Data

ID		Inputs		Target
Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...

"All models are wrong, but some are useful"



all models are wrong (Box 1976)

You might be familiar with the phrase "All models are wrong, but some are useful" attributed to George Box. Well, in an era of massively abundant data, we don't have to settle for wrong models.

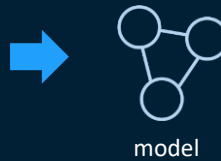
Let's begin our discussion of modeling with an overview of common modeling vocabulary. Modeling starts with a set of *analysis data* called *training data*. The observations in a training data set are known as *training cases* or simply *cases*. The variables are called *inputs* and *targets*. The measurement scale of the inputs and the target can be varied. The inputs and the target can be numeric variables, such as **age**, **hours per week**, and **income**. They can be nominal variables, such as **gender**.

Modeling Vocabulary

Model

A **model** is a concise representation of the inputs and target association.

Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...

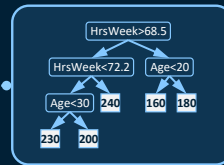


The purpose of the analysis data is to generate a model. A *model* is a concise representation of the relationship between the inputs and the target variables. For example, you might want to create a model to explain variability in income using age, gender, and weekly work hours, or you might want to predict income using these features. As such, a model is a formal exemplification of a theory and is usually specified as a mathematical relationship to approximate the reality.

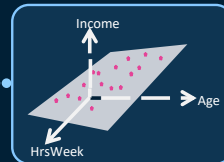
Modeling Vocabulary

Model Frameworks

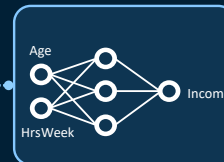
Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...



a decision
tree model



a regression
model

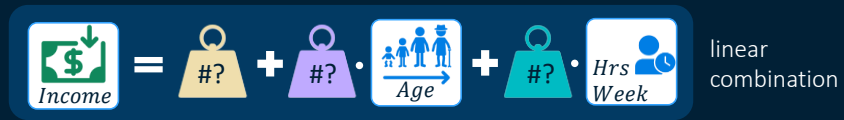


a neural
network model

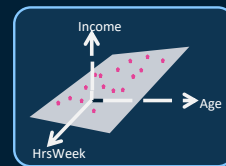
A model stores a representation of an input–output relationship in many frameworks. It could be a mathematical equation involving parameter estimates in a typical regression model, or a more complex mathematical equation involving weights and biases in a neural network model, or it might have certain prediction rules representing input–output relationships like in a decision tree model.

Modeling Vocabulary

Regression Model Example



Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...

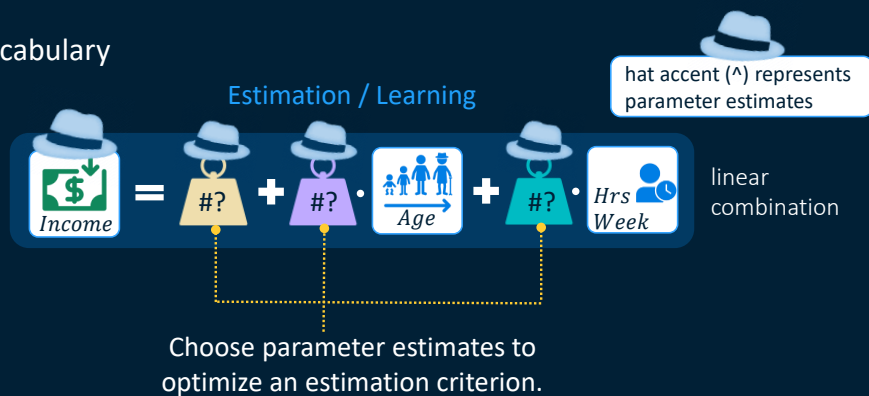


a regression model

Let's understand more modeling vocabulary using a simple example of a regression model. Linear regressions are usually deployed for targets with an interval measurement scale, like income as shown here with two inputs – age and weekly working hours. A prediction estimate for the target variable is formed from a linear combination of the inputs. Each input is weighted differently to determine the trend strength between each input and the target. A bias weight is also incorporated that centers the range of predicted target values. These are called model parameters and they are unknown.

You will learn about regression models in detail in one of the upcoming lessons.

Modeling Vocabulary



- In traditional statistical terms, it's called *estimation*.
- In a modern machine learning world, it's called *learning*.

The primary task in a regression model is to obtain parameter estimates. Statisticians generally represent parameter estimates using a hat accent. The estimates are best chosen by optimizing an estimation criterion represented mathematically. In traditional statistical terms, this process is called *estimation*, and in a more modern machine learning world, it's called *learning*. The bottom line is that the parameters are estimated or learned to optimize a mathematical function. The function you want to optimize is called the objective function, or estimation criterion.

Modeling Vocabulary

Residuals / Errors

$$\text{Income} = 5.15 + 0.66 \cdot \text{Age} + 4.22 \cdot \text{HrsWeek}$$

Name	Age	Gender	HrsWeek	Income
John	24	M	60	241.89
Smith	18	M	30	162.31
Emma	31	F	45	220.04
...	...	...	...	...



Income
274.19
143.63
215.51
...

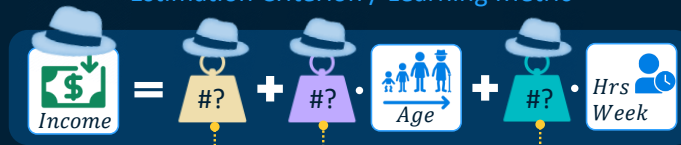
Residual / Error:

$$\text{Income} - \text{Income}$$

After the parameters are estimated, predicted values on the analysis data can be generated, such as predicted income. The difference between observed and predicted values of data are called *residuals* in a statistical or machine learning model. They are a diagnostic measure used when assessing the quality of a model. They are also known as *errors*.

Modeling Vocabulary

Estimation Criterion / Learning Metric



Choose parameter estimates to optimize an estimation criterion.

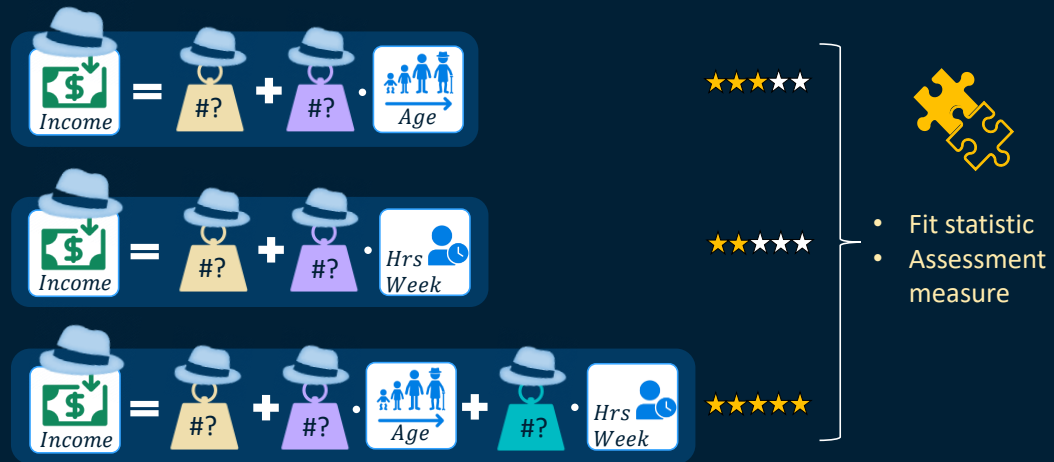
Minimize squared error function:
$$\sum \left[\text{Income} - \text{Income} \right]^2$$

At its core, the estimation criterion or learning metric is incredibly simple: It's a method of evaluating how well your algorithm models your data set.

For example, in a linear regression model, *parameter estimates* are chosen to minimize the squared error between the predicted and observed target values. This method is called *least squares estimation*. The prediction estimates can be viewed as a linear approximation to the expected (average) value of a target conditioned on observed input values.

Modeling Vocabulary

Model Fitting / Model Training



Fitting a model to data means choosing a statistical model that predicts values as close as possible to the ones observed in your population. In the machine learning world, determining the best fit model is also known as *model training*.

For example, several regression models can be created with multiple combinations of input variables. The purpose of model training is to build the best mathematical representation of the relationship between inputs and a target. While doing such a statistical analysis, it's very important to make sure about the goodness of fit of the model used. To do so, there are different *fit statistics* that you can use, such as R-square, the coefficient of determination. Fit statistics are generally called *assessment measures* in machine learning terminology.

Modeling Vocabulary

Predictions / Scores

$$\text{Income} = 5.15 + 0.66 \cdot \text{Age} + 4.22 \cdot \text{HrsWeek}$$

NEW

Name	Age	HrsWeek
Robert	45	50
Mary	21	40
James	34	22
...	...	...



Predicted Income
245.85
187.81
120.43
...

Predictions are output of the model given a set of input measurements.

When the *best-fit model* is determined, *predictions* can be generated. The outputs of the predictive model are referred to as *predictions*. Predictions represent your best guess for the target given a set of input measurements. The predictions are based on the associations learned from the training data by the predictive model. Predictions are generally called *scores* in machine learning terminology.

Question: Modeling Vocabulary (Model)

Which of the following defines the analytical term *model*?

- a. Choosing model parameter estimates by optimizing an estimation criterion
- b. A measure of evaluating how well your algorithm models your data set
- c. A concise representation of the inputs and target association
- d. A person with a role either to promote, display, or advertise commercial products

Question: Modeling Vocabulary (Model)

Which of the following defines the analytical term *model*?

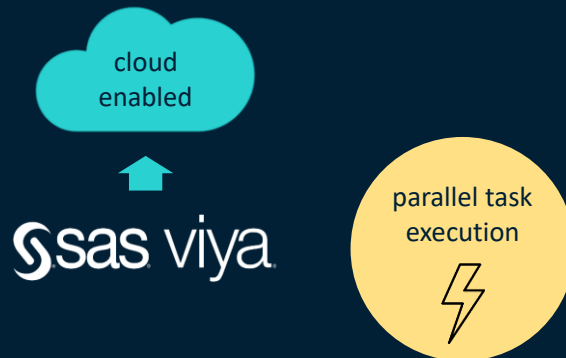
- a. Choosing model parameter estimates by optimizing an estimation criterion
- b. A measure of evaluating how well your algorithm models your data set
- ☒ c. A concise representation of the inputs and target association
- d. A person with a role either to promote, display, or advertise commercial products

The correct answer is **c**. A model is a concise representation of the relationship between the inputs and the target variables. Option **a** is the definition of estimation/learning. Option **b** is the definition of estimation criterion. Option **d** is irrelevant in analytical vocabulary

1.3 Introduction to SAS Viya and SAS Studio

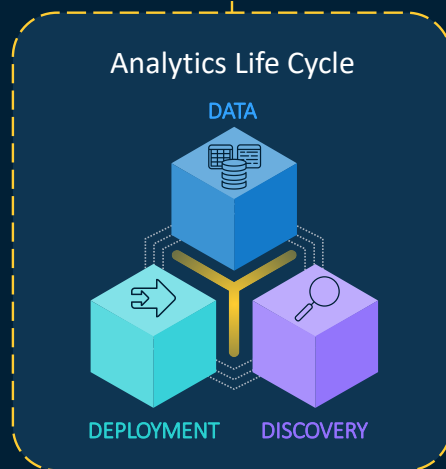


Introduction to SAS Viya



SAS Viya is the latest enhancement of SAS, which is cloud-ready analytics and data management platform. It allows scalable, web-based access for your data processing needs.

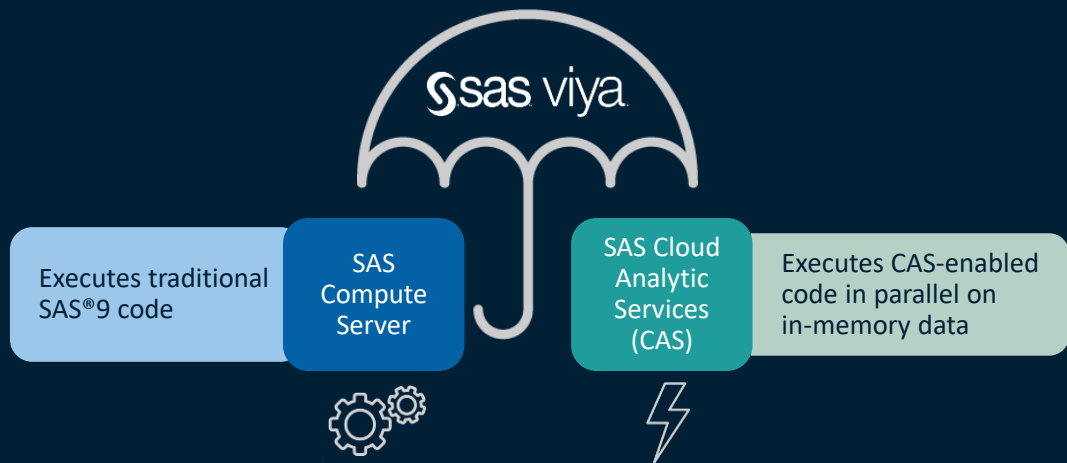
SAS Viya offers several products that facilitate parallel task execution. Most computers can execute operations in parallel due to their multicore infrastructure. Performing more than one operation simultaneously has the potential to speed up most tasks and has many practical uses within the field of data science.



Big data demands sophisticated data management and advanced analytics techniques. You can orchestrate your entire analytics journey – from data to tangible results – with a powerful analytics platform.

The "heart" of SAS is the analytics life cycle. The three phases of the analytics life cycle are data, discovery, and deployment. Data are the foundation of everything you do, discovery is the act of finding something that you had not known before, and deployment is where you get the value out of analytics.

SAS Viya Servers



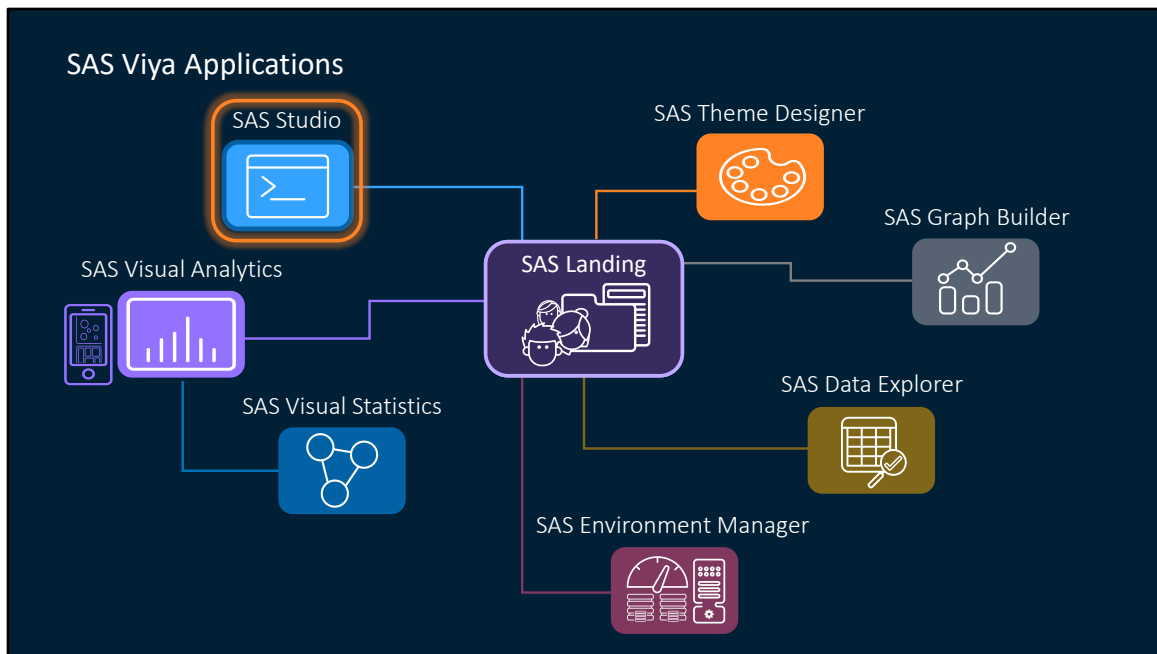
SAS Viya includes multiple servers to execute SAS code. The two primary servers are the SAS Compute Server and SAS Cloud Analytic Services, or CAS. You can think of the SAS Compute Server as equivalent to the SAS®9 workspace server. Your familiar SAS®9 programs can be submitted as is to SAS Viya and, by default, they will run on the Compute Server. There is no need to learn new syntax to use the Compute Server.

SAS Cloud Analytic Services is the high-performance server that performs parallel processing on in-memory data. It is likely that you will use CAS for your big data and complex analytics. Often, only very minor code modifications are required for programs to run in CAS.

More about SAS Viya

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.





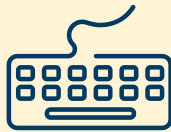
SAS Viya consists of many different applications that enable you to take advantage of the powerful CAS server. The entry point for SAS Viya is SAS Landing, a collaborative interface for accessing, organizing, and sharing content. This application is the gateway to all other applications in Viya.

Our focus will be SAS Studio in this course.

SAS Studio

- SAS Studio is a web-based development interface which offers multiple methods for working with SAS code.

Traditional Programming



Program Editor

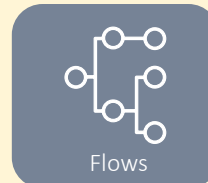


SAS Programs

SAS Studio Flows



Point-and-click



Flows

SAS Studio is a web-based development interface which offers multiple methods for working with SAS code.

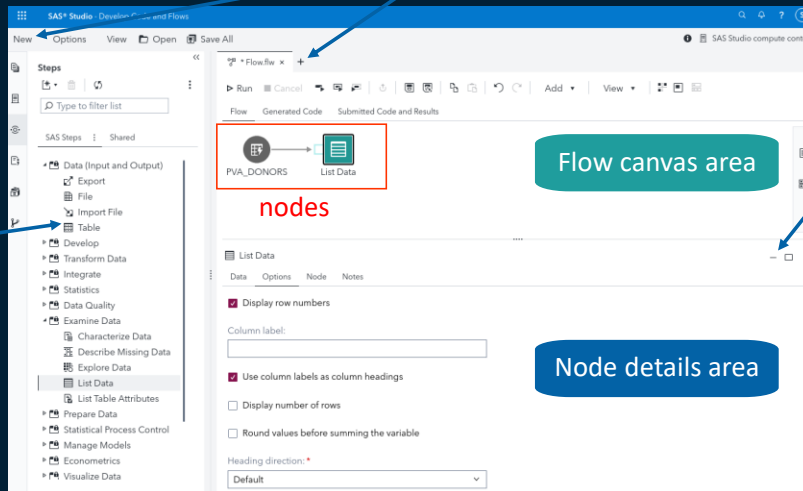
The traditional method is to use the program editor to write SAS code and develop SAS programs.

Another method is to use pre-built point-and-click steps which generate SAS code for tasks such as accessing, analyzing, transforming, or visualizing data. These steps are connected in a dynamic flow chart, allowing you to visualize your data transformation process.

SAS Studio flow example

Add a new program or flow

Add steps to a flow



Minimize/
maximize
details

Node details area

Here is an example flow. A new flow can be created using the plus sign to the right of an open tab or by using the New menu at the top left side of SAS Studio.

A flow is a sequence of operations on data. Data and operations are represented in SAS Studio by steps that can be added to a flow by expanding the Steps menu in the Navigation pane on the left and double clicking on individual steps. The steps will appear as nodes in the Flow canvas area in the center of the open flow tab. The current flow has 2 nodes: a table node labeled PVA_DONORS connected to a List Data node. Options for each node can be set in the Node details area through the point and click interface. The details can be minimized and maximized using the buttons on the upper right side of the details pane.

More information about constructing and executing SAS Studio flows will be shown in the course demonstrations

SAS Studio Flow steps

SAS Step Categories		
Data (Input and Output) ★	Enrichment	Prepare Data ★
Develop (Code)	Econometrics	Examine Data ★
Transform Data	Data Quality	Statistical Process Control
Integrate	Visualize Data	Text Analytics
Statistics ★	Machine Learning ★	Custom Steps
Manage Models	Optimization and Network Analysis	

Flow steps are organized into categories based on the user's desired task.

Specific steps are available to:

- Work with input or output data
- Develop code
- Transform data
- Integrate data
- Generate statistics
- Manage models
- Enrich data
- Work with econometrics
- Enhance data quality
- Visualize data
- Manage machine learning
- Perform optimization and network analysis
- Prepare data
- Examine data
- Execute statistical process control, or
- Complete text analytics

Additionally, SAS Studio offers a separate interface for users to design completely custom steps based on their business needs. Steps within the starred categories will be used to construct flows throughout this course.

Question: SAS Viya

SAS Viya is a cloud-ready analytics and data management platform that facilitates parallel task execution.

- a. True
- b. False

Answer: SAS Viya

SAS Viya is a cloud-ready analytics and data management platform that facilitates parallel task execution.

- ☒ a. True
- ☐ b. False

Answer: True

SAS Viya is the latest enhancement of SAS, which is a cloud-ready analytics and data management platform. SAS Viya offers several products that facilitate parallel task execution.